

$$\begin{aligned}\text{Now, } (ace)^{\frac{1}{p}} &= (b^p d^{2p} f^{3p})^{\frac{1}{p}} \\ &= (bd^2 f^3)^p \\ &= bd^2 f^3\end{aligned}$$

8. Option (c) is correct.

Given that $-\sqrt{2}$ and $\sqrt{3}$ are roots of the given equation. Therefore, $\sqrt{2}$ and $-\sqrt{3}$ are also roots of the given equation.

$$\begin{aligned}\therefore x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0 \\ &= (x + \sqrt{2})(x - \sqrt{2})(x - \sqrt{3})(x + \sqrt{3}) \\ &= (x^2 - 2)(x^2 - 3) \\ &= x^4 - 5x^2 + 6 \\ \therefore a_3 &= 0, a_2 = -5, a_1 = 0 \text{ and } a_0 = 6\end{aligned}$$

9. Option (c) is correct.

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$$\text{Now, } \left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$$

$$\begin{aligned}\Rightarrow |z_1 + z_2| &= |z_1 - z_2| \\ \Rightarrow |(x_1 + x_2) + i(y_1 + y_2)| \\ &= |(x_1 - x_2) + i(y_1 - y_2)| \\ \Rightarrow (x_1 + x_2)^2 + (y_1 + y_2)^2 &= (x_1 - x_2)^2 + (y_1 - y_2)^2 \\ \Rightarrow x_1^2 + x_2^2 + 2x_1x_2 + y_1^2 + y_2^2 + 2y_1y_2 \\ &= x_1^2 + x_2^2 - 2x_1x_2 + y_1^2 + y_2^2 - 2y_1y_2 \\ \Rightarrow x_1x_2 + y_1y_2 &= 0 \quad \dots(i) \\ \frac{z_1}{z_2} &= \frac{x_1 + iy_1}{x_2 + iy_2} \times \frac{x_2 - iy_2}{x_2 - iy_2} \\ &= \frac{(x_1x_2 + y_1y_2) + i(y_1x_2 - x_1y_2)}{x_2^2 + y_2^2} \\ &= 0 + \frac{i(y_1x_2 - x_1y_2)}{x_2^2 + y_2^2}\end{aligned}$$

$$\therefore \operatorname{Re}\left(\frac{z_1}{z_2}\right) = 0$$

$$\text{Now, } \operatorname{Re}\left(\frac{z_1}{z_2}\right) + 1 = 1$$

10. Option (b) is correct.

$$\therefore 26! = n8^k = n2^{3k}$$

Maximum power of 2

$$\begin{aligned}&= \left[\frac{26}{2} \right] + \left[\frac{26}{4} \right] + \left[\frac{26}{8} \right] + \left[\frac{26}{16} \right] \\ &= 13 + 6 + 3 + 1 = 23\end{aligned}$$

But power of 2 is $3k$

$$\therefore \text{Maximum value of } 3k = 21$$

$$\Rightarrow k = 7$$

11. Option (b) is correct.

$$(1) \therefore \operatorname{adj}(AB) = (\operatorname{adj} B)(\operatorname{adj} A)$$

$\therefore$ Statement 1 is wrong.

$$(2) \therefore AB \neq BA$$

$$\therefore \operatorname{adj}(AB) \neq \operatorname{adj}(BA)$$

$\therefore$ Statement 2 is wrong.

$$(3) (AB) \operatorname{adj}(AB) - |AB| I_n$$

$$\Rightarrow |AB| I_n - |AB| I_n \quad [\because A \operatorname{adj} A = |A| I]$$

$$= \text{Null matrix}$$

$\therefore$ Statement 3 is correct.

12. Option (d) is correct.

By properties statement 1 is correct.

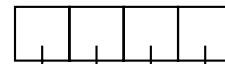
We know that If $A^n = A$ then A is identity matrix.

$\therefore$ Statements 2 and 3 are correct.

So, all statements are correct.

13. Option (b) is correct.

Even digit = 0, 2, 4, 6, 8.



Choice $\rightarrow$ 4 5 5 5

$$\therefore \text{Total required number} = 4 \times 5 \times 5 \times 5 = 500$$

14. Option (b) is correct.

$$(z - 100)^3 + 1000 = 0$$

$$\Rightarrow (z - 100)^3 = (-10)^3$$

$$\therefore z - 100 = -10 \quad (w, w^2, 1)$$

$$\therefore z - 100 = -10$$

$$\Rightarrow z = 90$$

$$z - 100 = -10w$$

$$\Rightarrow z = 100 - 10w$$

$$z - 100 = -10w^2$$

$$\Rightarrow z = 100 - 10w^2$$

15. Option (d) is correct.

$$\begin{aligned}(1 + i)^4 + (1 - i)^4 &= [(1 + i)^2]^2 + [(1 - i)^2]^2 \\ &= (1 - 1 + 2i)^2 + (1 - 1 - 2i)^2 \\ &= (2i)^2 + (-2i)^2 \\ &= 4i^2 + 4i^2 \\ &= -4 - 4 = -8\end{aligned}$$

16. Option (a) is correct.

We know that all diagonal elements of skew symmetric matrix is zero then their sum is also zero.

So, statements 1 and 2 are correct.

We know that If $AA^T = I$ then

A is called orthogonal matrix.

$$\text{But } AA^T = A(-A) = -A^2 \quad [\because A^T = -A]$$

So, Statement 3 is wrong.